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QUARTERLY EXAMINATION - 2025

CLASS : 11 **PHYSICS** Reg.No :

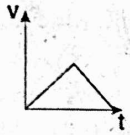
TIME : 3.00 Hrs. MARKS : 70

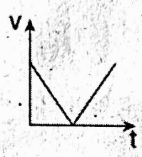
PART – I

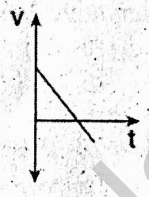
$15 \times 1 = 15$

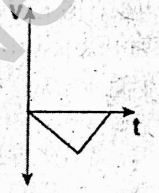
Note: (i) Answer all the questions.
(ii) Choose the most appropriate answer from the given four alternatives and write the option code and the corresponding answer

1. The velocity of a particle v at an instant t is given by $v = at + bt^2$. The dimensions of b is
   a) [L] b) $[LT^{-1}]$ c) $[LT^{-2}]$ d) $[LT^{-3}]$
2. When a car takes a sudden left turn in the curved road, passengers are pushed towards the right due to
   a) inertia of direction b) inertia of motion c) inertia of rest d) absence of inertia
3. Escape velocity on Earth is 11.2 km s^{-1}. What would be the escape velocity on a planet whose mass is 1000 times and radius is 10 times that of Earth?
   a) 112 km s^{-1} b) 11.2 km s^{-1} c) 1.12 km s^{-1} d) 3.7 km s^{-1}
4. A ball is projected vertically upwards with a velocity v. It comes back to ground in time t. Which v-t graph shows the motion correctly?

   a) 

   b) 

   c) 

   d) 

5. What is the minimum velocity with which a body of mass m must enter a vertical loop of radius R so that it can complete the loop?
   a) $\sqrt{2gR}$ b) $\sqrt{3gR}$ c) $\sqrt{5gR}$ d) $\sqrt{gR}$
6. In an elastic collision
   a) momentum and kinetic energy are conserved b) only momentum is conserved
   c) only kinetic energy is conserved d) momentum and kinetic energy are not conserved
7. If the length and time period of an oscillating pendulum have errors of 1% and 3% respectively then the error in measurement of acceleration due to gravity is
   a) 4% b) 5% c) 6% d) 7%
8. If a person moves from Chennai to Trichy, his weight
   a) Increases b) Decreases
   c) remains same d) increases and then decreases
9. A body of mass 2 kg travels according to the law $x = pt + qt^2 + rt^3$, where $q = 4\ ms^{-2}, p = 3\ ms^{-2}$ and $r = 5\ ms^{-2}$. The force acting on the body at $t = 2\ s$ is
   a) 136 N b) 134 N c) 158 N d) 68 N
10. A vehicle starts from rest and accelerates uniformly to a speed of $180\ km\ h^{-1}$ in $10\ s$. The distance covered by the vehicle in the time interval is
    a) 200 m b) 300 m c) 500 m d) 250 m
11. A couple produces,
    a) pure rotation b) pure translation
    c) rotation and translation d) no motion
12. If the linear momentum is increased by 50%, then kinetic energy will be increased by
    a) 50% b) 100% c) 125% d) 25%
13. An object of mass m begins to move on the plane inclined at an angle θ. The coefficient of static friction of incl surface is μ_s. The maximum static friction experienced by the mass is
    a) mg b) $\mu_s mg$ c) $\mu_s mg \sin\theta$ d) $\mu_s mg \cos\theta$
14. According to Kepler's second law, the radial vector to a planet from the Sun sweeps out equal areas in equal intervals of time. This law is a consequence of
    a) conservation of linear momentum b) conservation of angular momentum
    c) conservation of energy d) conservation of kinetic energy
15. Three measurements are made as 18.425 cm, 7.21 cm and 5.0 cm. The addition should be written as
    a) 30.635 cm b) 30.64 cm c) 30.63 cm d) 30.6 cm

Kindly Send Me Questions & Answer Keys to Us: padasalai.net@gmail.com

PART – II

$6 \times 2 = 12$

Answer any six questions. Question no. 24 is compulsory:

16. How will you measure the diameter of the Moon using parallax method?
17. Consider two masses of 10 g and 1 kg moving with the same speed 10 m s^{-1}. Calculate the magnitude of the momentum.
18. Show that impulse is the change of momentum.
19. Write the various types of potential energy.
20. Give any two examples of torque in day-to-day life.
21. Why is the energy of a satellite (or any other planet) negative?
22. Define a vector. Give examples
23. State the empirical laws of static and kinetic friction.
24. A box is pulled with a force of 25 N to produce a displacement of 15 m. If the angle between the force and displacement is 30°, find the work done by the force.

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PART – III

$6 \times 3 = 18$

Answer any six questions. Question no. 33 is compulsory:

25. Explain the principle of homogeneity of dimensions. Give example?
26. A spider of mass 50 g is hanging on a string of a cob web. What is the tension in the string?
27. What is the difference between velocity and average velocity?
28. Write the differences between conservative and Non-conservative forces. Give 2 examples each.
29. Find out the centre of mass for the given geometrical structures. (1) Equilateral triangle (2) Cylinder (3) Square
30. State Kepler's three laws.
31. Using free body diagram, show that it is easy to pull an object than to push it.
32. Explain the characteristics of elastic and inelastic collision.
33. Find the moment of inertia of a disc of mass 3 kg and radius 50 cm about the axis touching the edge and perpendicular to the plane of the disc

PART – IV

$5 \times 5 = 25$

Answer all the questions:

34. a) State and prove parallel axis theorem?

(Or)

b) What do you mean by propagation of errors? Explain the propagation of errors in addition and multiplication?

35. a) Explain the motion of blocks connected by a string in vertical motion

(Or)

b) Discuss the properties of scalar products.

36. a) Derive an expression for escape speed?

(Or)

b) Explain the method to find the centre of gravity of an irregularly shaped lamina?

37. a) Derive the expression for centripetal acceleration.

(Or)

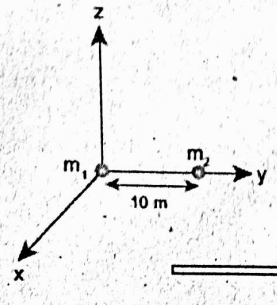
b) State and Explain the Work – Kinetic energy theorem?

38. a) If the value of universal gravitational constant in SI is 6.6×10^{-11} N m^2 kg^{-2}, then find its value in CGS System?

(Or)

b) (i) State Newton's Universal law of gravitation.

(ii) Consider two point masses m_1 and m_2 which are separated by a distance of 10 meter as shown in the following figure. Calculate the force of attraction between them and draw the directions of forces on each of them. Take $m_1 = 1$ kg and $m_2 = 2$ kg



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